

SARTHAK SHINDE

AI/ML Engineer — Pune, Maharashtra, India

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Professional Summary

Computer Science graduate specializing in Artificial Intelligence, Machine Learning, and Computer Vision. Built and deployed a production Machine Learning and OCR pipeline at Tata Motors that replaced a fully manual document workflow, now used by 10+ stakeholders. Hands-on experience across the full AI/ML lifecycle, including data preprocessing, model training, Generative AI (RAG), API development, and cloud-ready containerized deployment. AWS Certified Cloud Practitioner, with hands-on cloud computing fundamentals. Seeking a Junior AI/ML Engineer, Data Scientist, or Generative AI role.

Technical Skills

Languages: Python, SQL

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Regression, K-Means Clustering, Feature Engineering, Model Evaluation, Model Deployment, Deep Learning

Computer Vision: OpenCV, Tesseract OCR, Image Preprocessing, Adaptive Thresholding

Generative AI & NLP: LangChain, Hugging Face, Prompt Engineering, RAG (Retrieval-Augmented Generation), Large Language Models (LLMs), Natural Language Processing (NLP)

Cloud & DevOps: AWS Cloud Practitioner Essentials (Certified), Docker, Git, CI/CD basics, REST APIs

Backend & Deployment: FastAPI, Flask, Microservices, API Development

Data Science: Pandas, NumPy, PostgreSQL, EDA (Exploratory Data Analysis), Statistical Analysis

Visualization: Streamlit, Plotly, Matplotlib

Work Experience

AI/ML & Data Science Intern

Oct 2025 – Jan 2026

Tata Motors

Pune, Maharashtra, India

- Built a Machine Learning and Computer Vision-based OCR pipeline with **OpenCV** and **Tesseract** to digitize 1,000+ scanned trainee assessment records, replacing a fully manual data entry process.
- Tuned a preprocessing chain (Gaussian denoising, adaptive thresholding, morphological transforms) that noticeably improved text extraction accuracy on low-quality scans.
- Deployed a Dockerized **FastAPI** microservice with REST APIs so 10+ stakeholders could view KPI dashboards directly instead of waiting on weekly Excel updates.
- Performed EDA, feature engineering, and statistical validation across training cohort data using **Pandas/NumPy** to help the team identify which factors most affected training outcomes.

Projects

No-Code ML Analytics Platform (AADES) | *Python, Streamlit, Pandas, Scikit-learn, Plotly* | [Live Demo](#)

- Built a no-code Machine Learning analytics tool that runs schema profiling, statistical visualization, K-Means clustering, and linear regression on any uploaded dataset.
- Added interactive Plotly visualizations (scatter matrices, residual plots, elbow curves) so non-technical users can explore results without writing code; deployed on Streamlit Cloud.

RAG Document Q&A System (Generative AI) | *Python, FastAPI, LangChain, Gemini, ChromaDB, Streamlit, Docker* | [Live Demo](#)

- Built a Retrieval-Augmented Generation (RAG) pipeline that lets users upload multiple PDFs and query them in natural language using a Large Language Model, returning answers with exact page-level citations.
- Used **LangChain** with a **ChromaDB** vector database for semantic retrieval, embeddings, and conversational memory across documents.
- Deployed as a Dockerized multi-service microservice architecture (FastAPI backend + Streamlit frontend) with a single-container endpoint for easy reproduction.

Market Sentiment & Trading Performance Dashboard | *Python, Scikit-learn, Streamlit, Plotly, NLP*

- Built an NLP and K-Means clustering pipeline that segments trading activity into behavioral archetypes (bullish, bearish, neutral) from sentiment and financial data.
- Built a Streamlit + Plotly dashboard surfacing correlations between sentiment and trading patterns in real time.

Education

Nutan College of Engineering & Research

July 2026

B.Tech in Computer Science & Engineering

Pune, Maharashtra, India — CGPA: 7.00

Certifications & Courses

- AWS Certified Cloud Practitioner — Amazon Web Services (2026)
- Generative AI with Large Language Models — Coursera (2025)
- Applied Machine Learning — Coursera (2025)
- Large Language Models — Coursera (2025)
- Data Science Essentials with Python — Cisco Networking Academy (2026)
- Python Programming — HackerRank (2025)